Physics 468/668: Statistical Physics **Spring 2019**

Lecture 3: The Statistical Basis of Thermodynamics

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3.1 The classical ideal gas

The key to derive the thermodynamic quantities is to solve S. We shall now derive the various thermodynamic properties of a classical ideal gas composed of monoatomic molecules. The main reason why we choose this high specialized system for consideration is that it affords an explicit, asympototic evaluation of the number $\Omega(N, V, E)$. In addition, this study serves as a useful reference with which the behavior of other physical systems, especially real gases, can be compared.

Again, we shall start with some assumptions.

1. noninteracting particles

2. distinguishable

$$\begin{aligned} \text{One particle:} \quad & \Omega(1, E, V) \propto V \\ \text{N particles:} \quad & \Omega(N, E, V) \propto V^N \end{aligned} \tag{3.1}$$

$$\frac{P}{T} = k\left(\frac{\partial \ln \Omega(N, E, V)}{\partial V}\right)_{N,E} = k\frac{N}{V} \tag{3.2}$$

If the system contains n moles of the gas, then $N = nN_A$, where N_A is the Avogadro number. The above equation can lead to ideal gas law.

$$PV = NkT = nRT \tag{3.3}$$

3.2 Intensive and extensive properties

Physical properties of materials and systems can often be categorized as being either *intensive* or *extensive* quantities, according to how the property changes when the size (or extent) of the system changes.

1. An *intensive* property is one whose magnitude is independent of the size of the system. (P,T)

2. An *extensive* property is one whose magnitude is additive for subsystems. (S, V)

3.3 Compute the Multiplicity

To derive other quantities, we need a detailed knowledge of the way Ω depends on the parameters N, V, E. The problem essentially reduces to determining the total number of ways satisfying the equation.

$$\sum_{r=1}^{3N} \epsilon_r = E \tag{3.4}$$

where ϵ_r are the energies associated with the various degrees of freedom of the N particles. For a molecule confined to the box, its wave-function must go to zero at each end, and therefore the allowed standing wave patterns are limited to the wavelengths of

$$\lambda_n = \frac{2L}{n}, \quad n = 1, 2, 3, ... \tag{3.5}$$

Thus

$$p_n = \frac{h}{\lambda_n} = \frac{hn}{2L} \quad n = 1, 2, 3, ... \tag{3.6}$$

Now the energy eigenvalues for a free, non-relativistic particle confined to a cubic box of side L, under the condition that the wave-function $\Psi(r)$ vanishes everywhere on the boundary are given by

$$\epsilon_r(n_x, n_y, n_z) = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2); \quad n_x, n_y, n_z = 1, 2, 3, ... \tag{3.7}$$

The number of distinct eigenfunctions should be equal to the number of independent, positive-integral solutions of the equation

$$(n_x^2 + n_y^2 + n_z^2) = \frac{8mV^{2/3}\epsilon}{h^2} = \epsilon^* \tag{3.8}$$

We may denote the number by $\Omega(1, \epsilon, V)$. Extending this argument, it follows that the desired number $\Omega(N, E, V)$ would be equal to the number of independent, positive integral solutions of the equation.

$$\sum_{r=1}^{3N} n_r^2 = \frac{8mV^{2/3}\epsilon}{h^2} = E^* \tag{3.9}$$

From the expression on the right side, we conclude that the **volume** V **and the energy** E of the system enter into the expression for Ω in the form of the combination $(V^{2/3}E)$.

$$S(N, V, E) = S(N, V^{2/3}E). \tag{3.10}$$

Hence, for the **consistency of** S **and** N, which defines a **reversible adiabatic process**,

$$V^{2/3}E = \text{const} \tag{3.11}$$

$$P = -\left(\frac{\partial E}{\partial V}\right)_{N,S} = \frac{2E}{3V} \tag{3.12}$$

Also, we shall get,

$$PV^{5/3} = \text{const} \tag{3.13}$$

which tells us how P varies with V during the reversible adiabatic process.

Question: how the relation was derived in thermodynamic class?

3.4 From the ideal gas picture

Suppose we have a single gas atom (Ar), with kinetic energy U in a container of volume V, what is its corresponding Ω? Obviously, the possible microstate is proportional to V. In principle, the atom can stay at any place of V. Also, each microstate can be represented as a vector, since it has velocity (more precisely Momentum). Therefore

$$\Omega \approx V \cdot V_p \tag{3.14}$$

It appears that both V and V_p somehow relate to very large numbers, would their product go to infinity? Fortunately, we have the famous **Heisenberg uncertainty principle**:

$$\Delta x \Delta p_x = h. \tag{3.15}$$

For a one-dimensional chain, we define L as the length in real space, L_p as the length in momentum space,

$$\Omega_{1D} = \frac{L}{\Delta x}\frac{L_p}{\Delta p_x} = \frac{LL_p}{h}. \tag{3.16}$$

Therefore, its 3D version is,

$$\Omega_1 = \frac{VV_p}{h^3}. \tag{3.17}$$

Accordingly, the multiplicity function for an ideal gas of two molecules should be

$$\Omega_2 = \frac{1}{2}\frac{V^2}{h^6} \times \text{area of } P \text{ hypersphere} \tag{3.18}$$

if the two molecules are indistinguishable. The general form for N should be

$$\Omega_N = \frac{1}{N!}\frac{V^N}{h^{3N}} \times \text{area of } P \text{ hypersphere}. \tag{3.19}$$

For N=1, how to calculate the area? Since E depends on the momentum by

$$E = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2) = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) \tag{3.20}$$

$$p_x^2 + p_y^2 + p_z^2 = \sqrt{2mE} \tag{3.21}$$

So the momentum space is the surface of a sphere with radius $\sqrt{2mE}$, namely,

$$\begin{aligned}
\text{area} &= 2 && (d\text{=}1)\\
&= 2\pi r && (d\text{=}2)\\
&= 4\pi r^2 && (d\text{=}3)\\
&= .... &&\\
&= .... &&\\
&= \frac{2\pi^{d/2}}{\Gamma(d/2)} r^{d-1} && (d \text{ in general})
\end{aligned} \tag{3.22}$$

Therefore, the general Ω is

$$\Omega_N = \frac{1}{N!}\frac{V^N}{h^{3N}}\frac{2\pi^{3N/2}}{(3N/2-1)!}\sqrt{2mE}^{3N-1}. \tag{3.23}$$

$$\Omega(U,V,N) = f(N)V^N E^{3N/2} \tag{3.24}$$

where $f(N)$ is a complicated function of N.

$$\ln\Omega \approx \ln\Sigma \approx N\ln\left[\frac{V}{h^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{3}{2}N \tag{3.25}$$

$$S(N,V,E) = k\ln\Omega = Nk\ln\left[\frac{V}{h^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{3}{2}Nk \tag{3.26}$$

which can be inverted to give

$$E(S,V,N) = \frac{3h^2N}{4\pi mV^{2/3}}\exp(\frac{2S}{3Nk} - 1) \tag{3.27}$$

The temperature of the gas then follows:

$$T = \frac{\partial E}{\partial S} = \frac{2}{3Nk}E \tag{3.28}$$

$$E = N(\frac{3}{2}kT) \tag{3.29}$$

$$C_V = (\frac{\partial E}{\partial T})_{N,V} = \frac{3}{2}Nk \tag{3.30}$$

$$C_P = (\frac{\partial E + PV}{\partial T})_{N,V} = \frac{5}{2}Nk \tag{3.31}$$

$$P = (\frac{\partial E}{\partial V})_{N,S} = \frac{2E}{3V} \tag{3.32}$$

The ratio between C_P and C_V is $\gamma = C_P/C_V = 5/3$.

The considerations of this section clearly demonstrated that the thermodynamics of a macroscopic system can be derived from the Ω. The whole problem then hinges on an asymptotic enumeration of these numbers, which unfortunately is tractable only for a few cases. But there remains an inadequacy that could not be detected in the derivations made so far, this relates to the explicit dependence of S and N.

3.5 Homework

1. prove $PV^{5/3}$ is a constant for ideal gas under adiabatic process.
2. 1.1a (do Taylor expansion for $\Omega(E)$ on $E - \bar{E}$)
3. 1.3 ($dS >= 0$)
4. 1.7 (see lecture)
5. 1.9 (express $S(N,V,E)$ as $Nf(V/N, E/N)$)